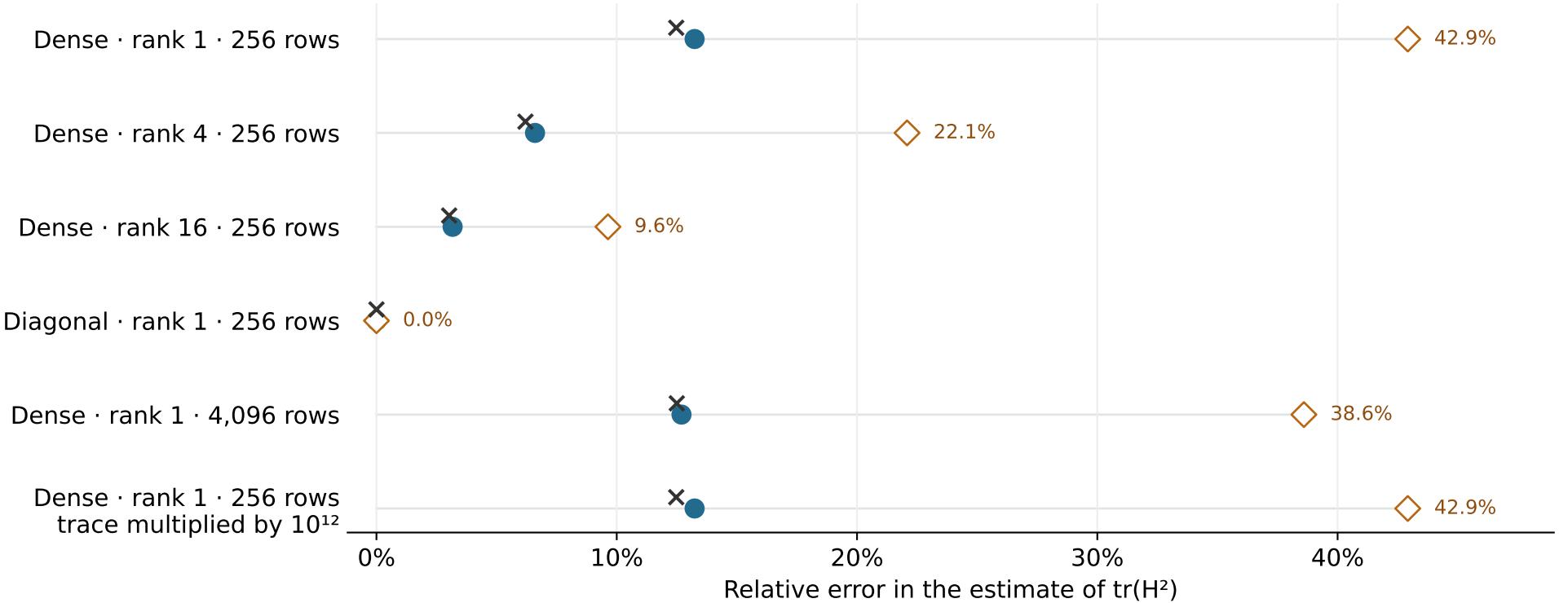
Large matrices can still have large probe error

128 probes · 256 seeds per operator · synthetic matrix constructions, not fitted survey datasets



RMS is a measure across random draws, not a per-run limit. The observed maximum is not a population bound.
Increasing dimension or trace magnitude does not repair a dominant dense direction; rank and orientation matter.